

Frontend assignment-2① CSS Box model

- every HTML element is treated as a rectangular box made of four layers, from inside out: content (text/images), padding, border & margin.

2. media query

- A media query applies CSS rules conditionally based on device characteristics (like screen width, height or orientation).

Ex:-

```
body {
  font-size: 16px;
}
media (max-width: 600px) {
  body {
    font-size: 12px;
    background-color: lightblue;
  }
}
```

③ Features of Bootstrap

1. Responsive 12 column grid system
2. Pre-styled components
3. Mobile first design approach
4. Cross-browser compatibility

- (y) Bootstrap grid structure example:-

→ The "Container" centers content,
"row" creates a horizontal group.
each (col-md-y) takes y of 12 grid columns.

→ Pattern used to select & target HTML elements for styling.

- 1) ~~element~~ Selector: $\{ \}$ = Selects all `<p>` tags
- 2) Class Selector: `.classname { }`
- 3) ID Selector: `#idname { }`
- 4) Universal Selector: `[type = "text"] { }`
- 5) ~~Selector~~ `~` : hover

6) CSS backgrounds

→ CSS backgrounds ~~are~~ Properties are used to Set The background color, image, position, and size of an element.

Properties

- background-color
- background-image
- background-repeat
- background-position
- background-size
- background-attachment

7) Overlap elements in CSS?

→ Elements can be overlapped Using Positioning and the z-index property.

example

```
.box 1 {
  position: absolute;
  left: 50px;
  top: 50px;
  z-index: 1;
}
```

```
.box 2 {
  position: absolute;
  left: 50px;
  top: 50px;
  z-index: 2;
}
```

Positioning Properties

- ⑧ 1) Static - default position
- ② relative - Positioned relative to its normal position.
- ③ absolute - Positioned relative to its nearest positioned ancestor.
- ④ fixed - Fixed relative to the browser window.
- ⑤ Sticky - behaves like relative until a specified scroll position.

Example

```

div {
  position: relative;
  top: 20px;
  left: 30px;
}

```

⑨ ECMAScript

- It is a standard / specification that defines how scripting languages such as JavaScript should work.
- JavaScript is an implementation of the ECMAScript Standard. It defines features such as:

- ① Variables
- ② Functions
- ③ Objects
- ④ Classes
- ⑤ Loops
- ⑥ Modules

⑩ Diff. Between var, let, & const.

Feature	var	let	const
Scope	Function	Block	Block
Re-declaration	Yes	No	No
Re-assignment	Yes	Yes	No
Hoisted	Yes	Yes	Yes
TDZ	No	Yes	Yes

TDZ (Temporal dead zone)

- It is the period between entering a block & the actual declaration of a 'let' or 'const' variable. During this period, the variable cannot be accessed.